

Heriverse-Heritage Science Metaverse

Development Service for a 3D online platform called
Heriverse-Heritage Science Metaverse for the collection, management, and
exploration of multimedial content for Heritage Science
as part of the PNRR Mission 4, “Istruzione e ricerca” –
Component 2, “Dalla ricerca all’impresa” – Investment line 3.1
“Fondo per la realizzazione di un sistema integrato di infrastrutture d
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H2IOSC”

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User Manual for “Heriverse-Heritage Science Metaverse” Platform

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Introduction

Heriverse – an acronym of Heritage Science Metaverse – is a concrete demonstration of how the latest European guidelines for cultural heritage can be translated into an operational environment already accessible to researchers, institutions, and professionals in the sector. Born as a “vertical pilot” within the Italian H2IOSC project, Heriverse combines three-dimensional exploration of objects and sites with a robust knowledge graph, addressing crucial issues such as multitemporality, traceability of data provenance, and scientific collaboration in an integrated manner.

A distinctive element of the platform is the semantic framework based on CIDOC-CRM, enhanced by the CIDOC-s3D extension and a data model formalized in JSON. This model, through the Python s3Dgraphy library, ensures interoperability and documentation rigor even for advanced 3D representations. This architecture, designed to power the future European Common Semantic Framework, allows for the connection of heterogeneous databases (from archeological surveys to sensor surveys) while maintaining transparency along the entire information chain.

In this way, Heriverse transcends the traditional “black box” of three-dimensional models, making the scientific origin of each data immediately accessible and fostering the convergence of academic publications, digital twins, and open data. The platform thus serves as a key hub for the Italian Digilab, promoting the sharing and dissemination of knowledge across the broader European cultural heritage ecosystem.

Main features

- 3D Scene Viewer:
 - View models organized by time period
 - Activating multiple graphs at once
 - Visualize the knowledge graph directly in the scene
 - Exploring the scene in VR
- Changing the composition of the graph:
 - Semantic enrichment by resources from different sources
 - Nodes and relations creation
 - Using the Shelf as a collection point for multimedia resources (local or remote) that can be integrated into existing graphs
 - Export the scene to modify it with external tools and then import it into the platform
- Collaborative mode:
 - Starting a shared work session
 - Communication via voice and/or text chat
 - Screen sharing capability

This manual guides the user step by step in using the platform, illustrating each feature in a practical way.

Access to the scene gallery

When the platform loads, the user interface presents a home screen containing a set of cards representing all the scenes marked as public by their creators. Each card serves as an access point to an interactive viewer, allowing exploration of the scene and the use of shared tools accessible to all users.

To access a scene viewer, simply select the relevant card: this activates the display of a detail sheet, from which you can launch the interactive component via a dedicated OPEN SCENE button.

Searching for a scene

In addition to the overview of public scenes, you can search for a scene by title. To do this, simply enter text in the bar below the title and click ENTER. The system will automatically display the scenes whose title contains the search string.

View options

By default, a grid layout is provided for displaying scene elements, with each element represented as a card. Using the buttons located immediately above the card section, you can also toggle between a list view of the elements. Furthermore, using the selector next to the pagination buttons, you can choose how many elements to display on a single page.

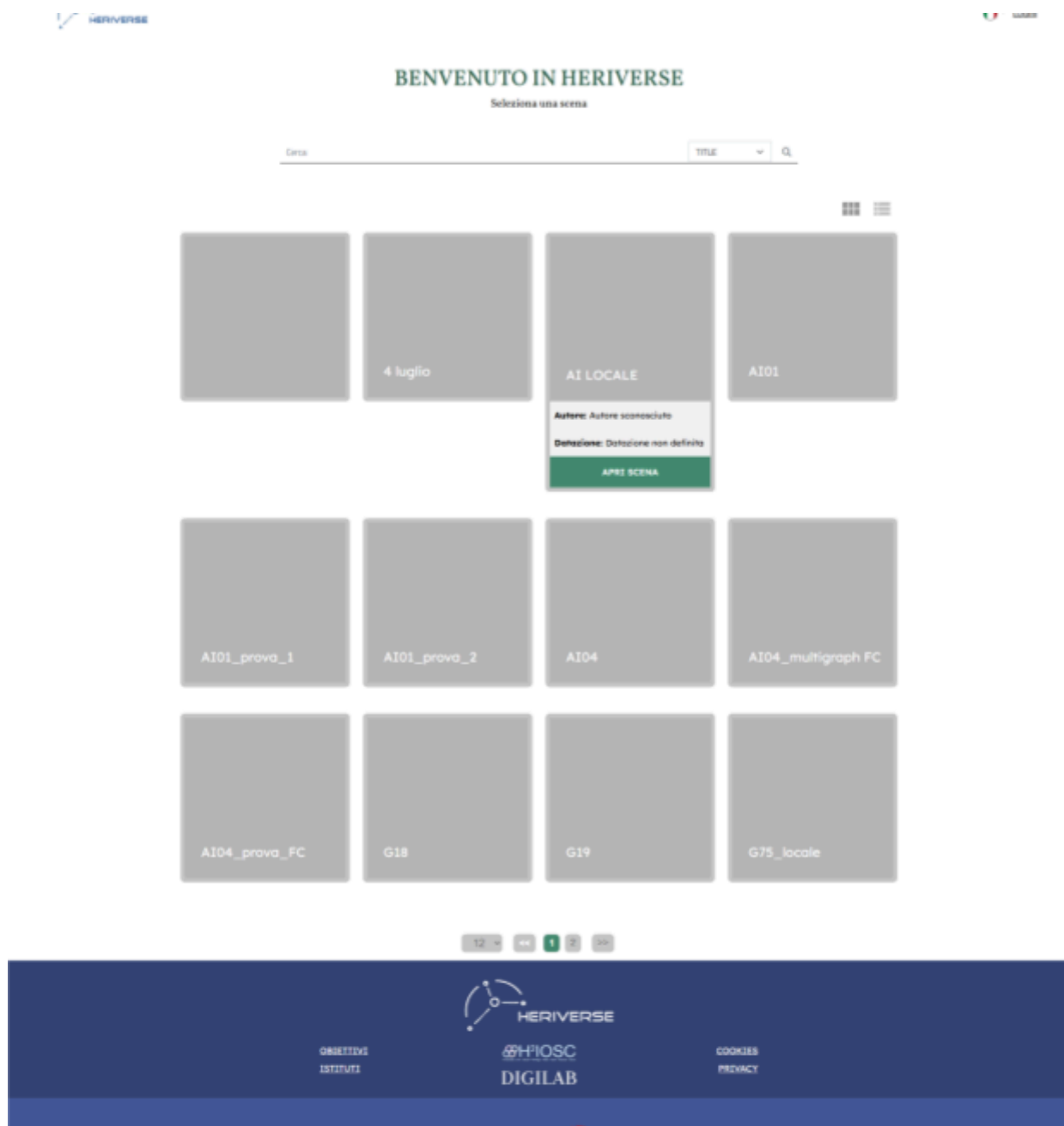


Figure 1: Scene page

Viewing a scene

Once you click the button that opens a scene's detail sheet, you access the viewer ([Figure 2](#)). The interface is divided into three main sections.

In the top bar, the scene title is visible in the center, followed immediately below by the active graph selector, which allows you to access multigraph mode. In this mode, you can filter the visible elements in the scene based on a selected time period. On the left side of the bar are two buttons: one allows you to return to the scene gallery, and the other starts VR mode (visible only if your device supports virtual reality). On the right side, there is another button that allows you to switch to editor mode, which is only active if the user has permissions to edit the scene; otherwise, you will be asked to log in.

The bottom bar contains tools for selecting and displaying elements in the scene, with behavior that varies depending on the active mode. When only one graph is present, a simple selector is displayed for switching between epochs. If multiple graphs are active, the bar is enriched with additional tools: a selector for choosing between the temporal filters already saved in the scene, two fields for manually entering the values of a custom filter, and a button to apply that filter ([Figure 2.1](#)). A switch, positioned before these tools, allows you to switch from this mode to an alternative selector that offers a series of filters automatically generated by the system.

The drop-down panel contains scene management controls. Here, you'll find a slider that adjusts the effect of ambient light on the displayed elements and a list of options to choose which relationships to display when viewing graphs. This customization enhances the exploratory experience within the scene.

While exploring, you may encounter semantic masks associated with some elements. Clicking on one of them displays a portion of the graph originating from the selected node. The nodes displayed are their direct neighbors, with the exception of "property" nodes, for which the portion of the graph extends to the associated document. Along with the graph drawing, a sidebar displays information about the selected node, including a list of its neighbors ([Figure 2.2](#)). By opening one of the neighbors' sections and clicking on a listed node, the sidebar automatically updates its content with information about the newly selected node, allowing for in-depth and dynamic navigation within the semantic network.

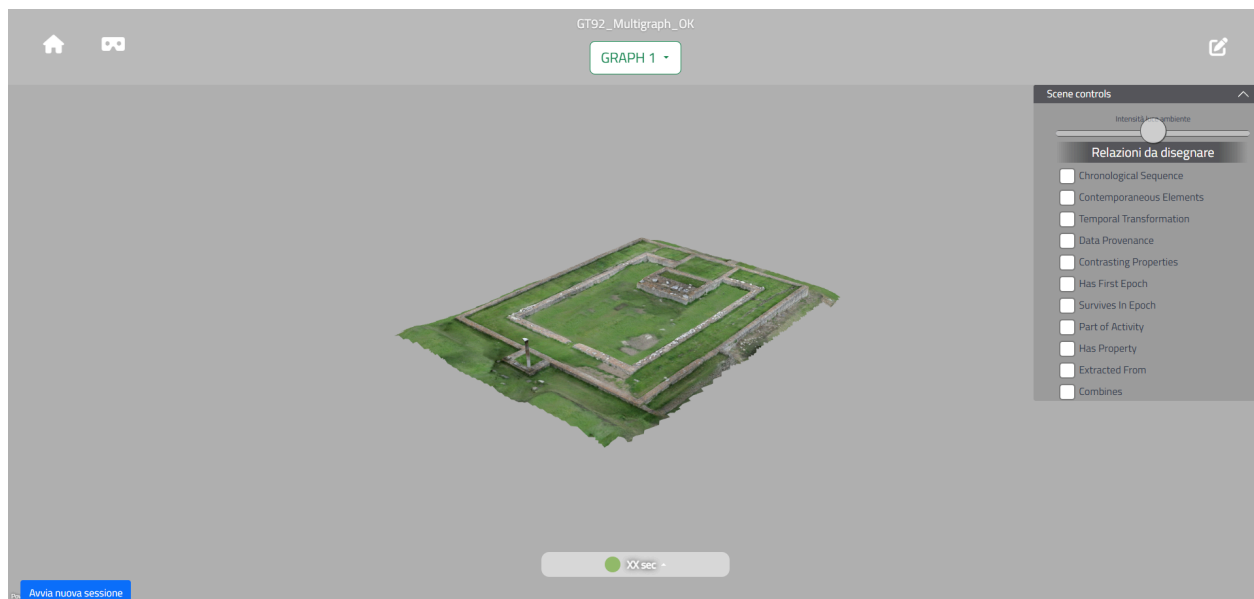


Figure 2: Scene viewer

Graph selector

Heriverse offers the ability to simultaneously display multiple graphs, if present in the scene's configuration JSON. Using the element under the title, you can choose which graphs to make active in the scene. Since at least one graph must be active, to activate another one separately, you must first select the desired option and then deselect the one previously displayed. When multigraphs are activated, the classic epoch selector (located at the bottom of the screen) gives way to the temporal filters section, which allows you to apply a filter to the nodes to display, not limited to the epochs relating to the individual graph.

Temporal filters

For multigraph mode, an alternative option is used to filter the nodes displayed in the scene. Temporal filters are used to scan the nodes of all active graphs to identify those to display. In the section ([Figure 2.1](#)), you can find:

- a selector, from which you can choose one of the filters previously saved by the scene creator;
- two forms in which you can enter values to create a custom filter;

Regardless of the mechanism chosen to define the filter, to apply its effect, you must click the FILTER button. The switch, located before the selector, allows you to activate an additional mode in which you can choose only filters created automatically by the system, allowing you to explore the contents of the graphs without having to worry about a time interval.



Figure 2.1: Temporal filter section

Interaction with the semantic shapes

Within the scene, it is possible to identify semantic shapes associated with specific elements in the environment. Their presence can be detected by moving the pointer around the scene and paying attention to the illuminated masks. Once the mask has been identified, you can click to draw the portion of the graph on the scene, with the node associated with the mask as its root and the nodes linked to it as neighbors according to the relationships selected from the scene controls panel. Along with the graph drawing, a side information panel is displayed ([Figure 2.2](#)).

Information panel

After interacting with a semantic shape, a panel appears on the right side of the screen containing various information about the selected node. It contains:

- Node name;
- Node description;
- Information about the author, license, and embargo date;
- Several sections relating to neighbors, grouped according to their relationship to the node in question.

Opening a neighbor's drop-down menu and clicking its contents will display the information for the clicked node in the side panel. Another way to access neighbor information is by clicking the corresponding nodes in the graph.



Figure 2.2: Graph visualization example

Scene Controls panel

Among all the elements present, this one in particular allows you to manage some settings related to the scene itself. In viewer mode, you can find:

- an ambient light slider, which allows you to adjust how much the model reflects the surrounding landscape;
- a list of checkboxes that allow you to choose which relationships to consider when drawing the graph.

Virtual Reality Mode

Using a VR headset, such as the Oculus Meta Quest, you can fully immerse yourself in the scene. After configuring the controls, virtual reality exploration can be started by pressing the dedicated button. Once in this mode, you can:

- Move using teleportation;
- Change epoch;
- Interact with semantic shapes;
- To exit this mode, simply press the **side trigger** on the **left controller**.

Movement in the scene

Movement within the environment is achieved through teleportation. Like other features, this can be done using controllers. To move, simply move the right controller to indicate the area you want to move to with the pointer. When a white circle appears at the end of the

radius, simply click the **front trigger** on the **right controller** to position yourself at the desired point.



Figure 2.3: Example of aiming for teleportation

Change of epoch

Another feature offered in this mode is the exploration of the various epochs associated with the graph. To select the desired epoch, simply use the panel on the left wrist (Figure 2.4). By default, this panel contains buttons that allow navigation between the various epochs. To access an epoch, simply point the **right controller** at the desired button and activate it by clicking the front trigger of the **right controller**. After this operation, the representation models associated with the selected epoch will appear.

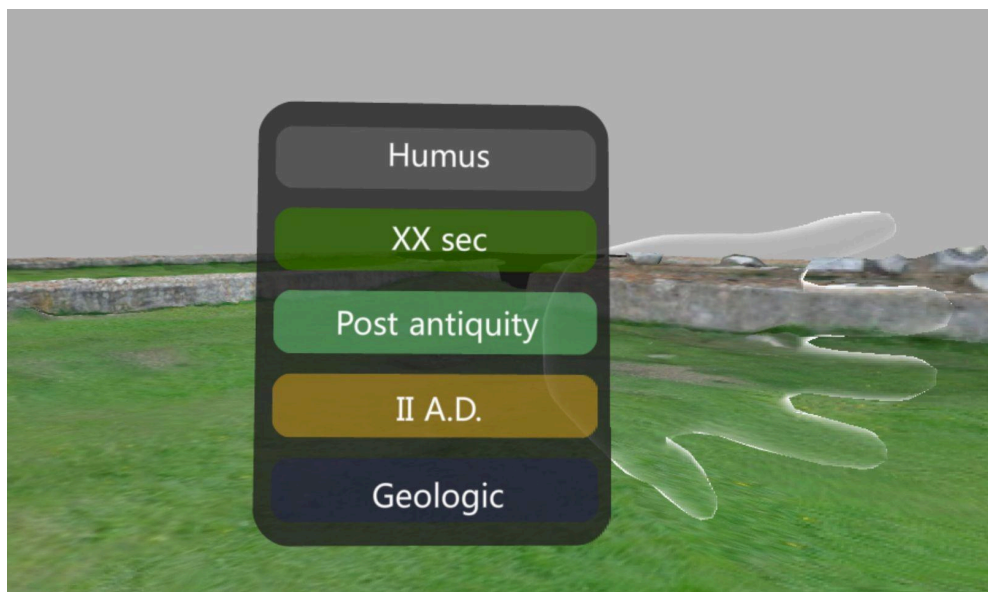


Figure 2.4: Example of epoch panel

Interaction with the environment

In addition to the two features described above, once you enter the scene, you can interact with the semantic shapes present. Using the **right controller**, simply orient it until the semantic shapes light up. Once you've identified the desired mask, simply click the **side trigger** on the **left controller** to draw the partial knowledge graph originating from the node to which the semantic shape belongs. As the graph appears at the clicked point, the content of the panel on the left wrist changes, replacing the epoch buttons with subpanels that summarize the nodes near the clicked one, grouping them based on their relationship to the selected node ([Figure 2.5](#)). The buttons at the bottom of the panel allow you to scroll through the various panels. Additionally, the name and description of the clicked node appear in the panel on the right wrist ([Figure 2.5](#)). After interacting with a semantic mask, you won't be able to interact with other masks. Pressing the side trigger on the left controller clears the drawn graph, resets the wrist panels to their default state, and lets you interact with the other masks.

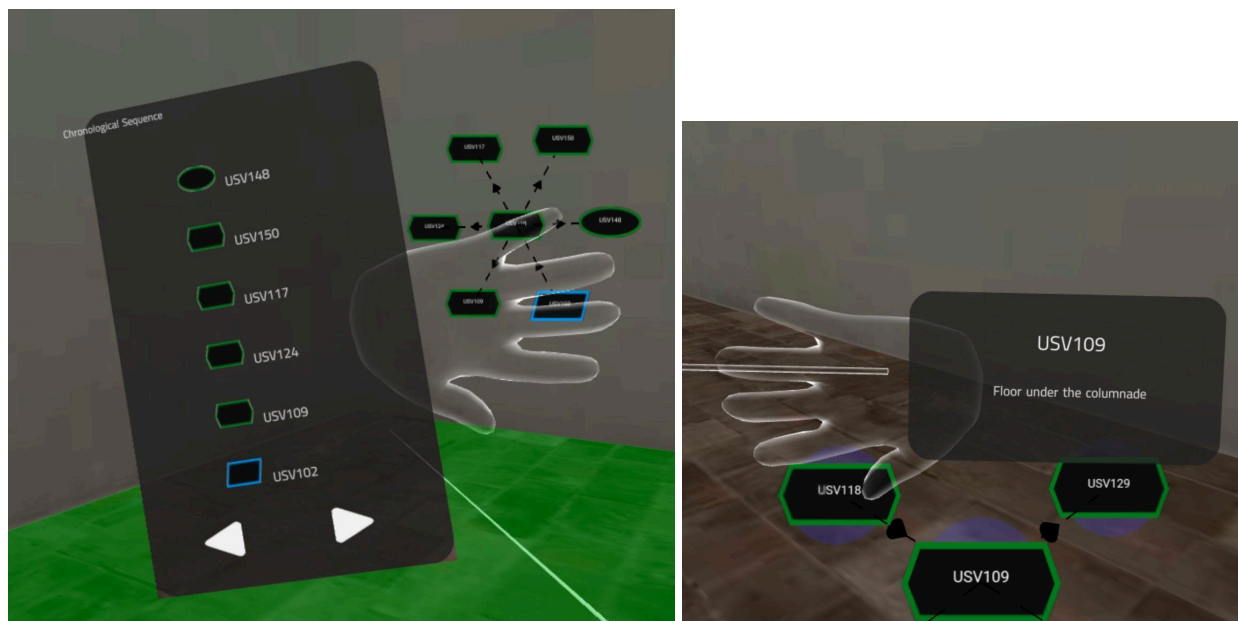


Figure 2.5: Example of a panel with a list of nodes (on the left) Example of a panel with the name and description of the clicked node

Authentication to the platform

To take advantage of the scene creation and editing features, the first thing to do is authenticate using the federated login offered by Digilab-IT. If you aren't logged in, you can reach the login section using the dedicated buttons at the top of the main page. From there, you'll be redirected directly to the Digilab authentication area ([Figure 3](#)), where you can also register a new user ([Figure 3.1](#)).

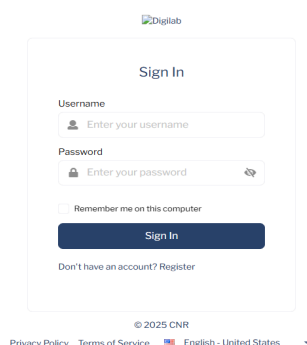
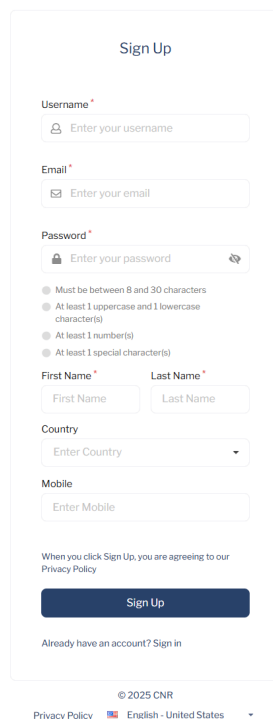
The image shows a web form titled 'Sign In' for Digilab. It includes a 'Username' field with a user icon and the placeholder 'Enter your username', and a 'Password' field with a lock icon and the placeholder 'Enter your password'. There is a 'Remember me on this computer' checkbox and a dark blue 'Sign In' button. Below the button is a link that says 'Don't have an account? Register'. At the bottom, there is a copyright notice '© 2025 CNR' and a footer with links for 'Privacy Policy', 'Terms of Service', and a language selector set to 'English - United States'.

Figure 3: Digilab Sign-In Screen



The image shows a 'Sign Up' form for Digilab. It includes fields for Username, Email, Password, First Name, Last Name, Country, and Mobile. The Password field has a strength indicator and a list of requirements: Must be between 8 and 30 characters, At least 1 uppercase and 1 lowercase character(s), At least 1 number(s), and At least 1 special character(s). Below the form is a 'Sign Up' button and a link for 'Already have an account? Sign in'. At the bottom, there is a copyright notice '© 2025 CNR' and a language selector 'English - United States'.

Figure 3.1: Digilab Sign-Up Screen

User Dashboard

After logging into Digilab, users can access Heriverse's creator features.

The first area available is the personal dashboard ([Figure 4](#)), where they can create a new scene or manage existing ones. This section, like the platform's main page, offers several features:

- Search scene by title
- Items shown in list or grid view
- Choose the number of items to display per page.

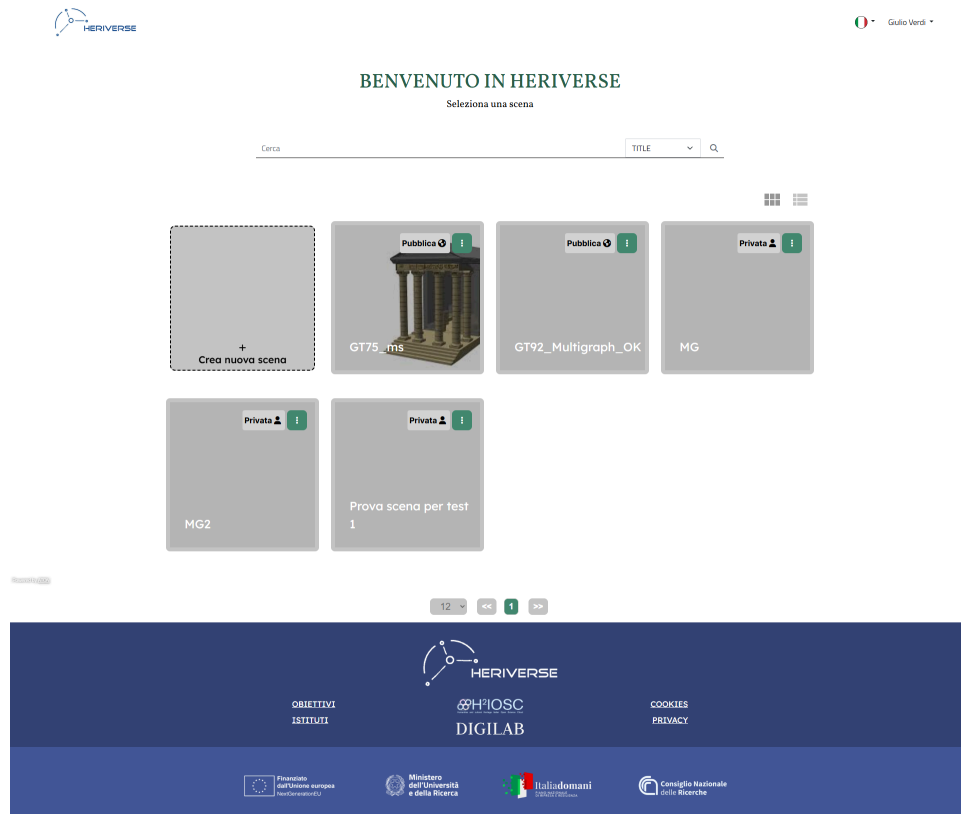


Figure 4: User dashboard

Creating a scene

To start using your own graph, you need to create a new scene ([Figure 4.1](#)) and import a ZIP file containing all the necessary elements.

When creating the scene, the first choice concerns visibility:

- set it to Public, if you want it to be accessible to all platform visitors;
- or to Private, to limit access to specific users.

Using the appropriate fields, you can also define who can view the scene and who is authorized to edit it.

In addition to the ZIP file containing the graph information, you can upload a panoramic image, which will be used as the background of the 3D environment.

HERIVERSE

PUBLIC
visible_to_all

PRIVATE
visible_to_coworkers

HERIVERSE

Italiano

English

Inserisci nome in Italiano

EDITORS

Seleziona editori

VIEWERS

Seleziona

DESCRIPTION

Italiano

English

Descrizione in Italiano

PREVIEW_IMAGE

Scegli file

Nessun file selezionato

AUTHOR

Autore

AGGIUNGI RISORSE

Panorama

Scegli file

Nessun file selezionato

File zip

Scegli file

Nessun file selezionato

SALVA SCENA

ANNULLA

Figure 4.1: Scene creation page

Editor mode

After creating the scene, you are automatically redirected to the editor page ([Figure 5](#)). In this section, in addition to the viewer tools, there are features dedicated to creating and editing nodes, relationships, and temporal filters (the latter useful in multigraph mode).

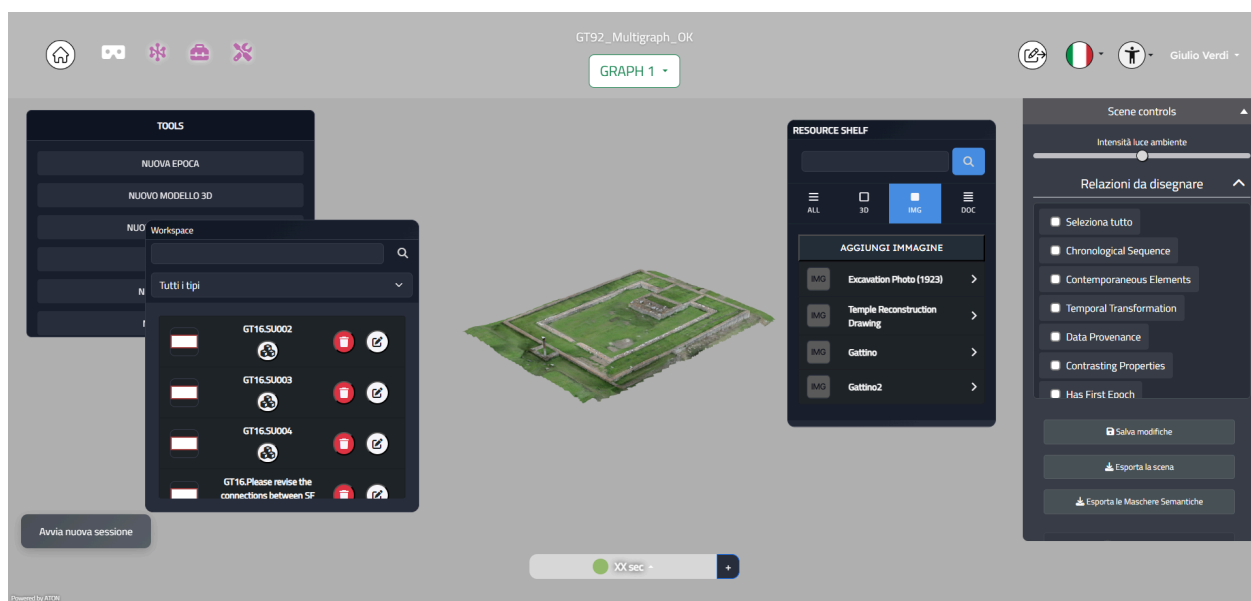


Figure 5: Editor mode screen

Creating a graph

Within the active graph selector, a button with a + appears and allow you to add an empty graph to the scene. Once clicked, a modal appears where you can enter the relevant information. By clicking **Done**, you can verify the presence of the new graph among existing ones.

Creating an epoch

Speaking of enriching the knowledge graph, one option is to create an epoch node. You can do this by clicking the button next to the epoch selector ([Figure 5.2](#)). After you click it, a modal dialog appears where you can enter:

- Epoch name;
- Epoch description;
- Start year;
- End year;
- Label color;
- Minimum height;

- Maximum height;

Since this button is available only when a single graph is active, you can also create an epoch node using the button in the Tools panel.

Temporal filters management

As described in the viewer section, when multiple graphs are active, the system switches from filtering by the epochs of the active graph to filtering items based on a predefined time filter. In addition to the controls described on the previous pages about the viewer, there are two additional buttons ([Figure 5.1](#)) that let you:

- Modify a temporal filter, which opens a modal dialog containing all the filter's details for editing.
- Save the filter; when values are entered in the text fields, a modal dialog appears where you can enter:
 - Filter name
 - Filter description
 - Label color

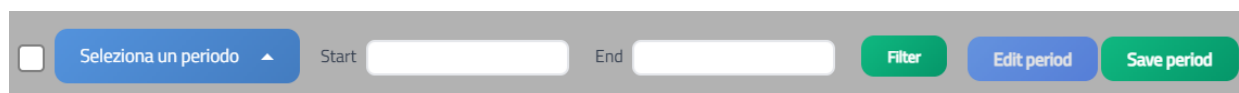


Figure 5.1: Temporal filters section in the editor



Figure 5.2: Epoch selector with button for creating new epochs

Semantic enrichment and node management

In editor mode, a set of panels is available to streamline operations. The key elements in this mode are:

- Workspace panel;
- Shelf panel;
- Tools panel.

Workspace panel

This component lists all the nodes in the active graphs, providing an overview of existing items and the ability to edit or remove them.

Node search

Given the potentially large number of nodes in the active graphs, a convenient tool is the search bar. To better orient yourself or find a specific node, simply start typing in the bar to filter the panel's content by name. Note that the search affects only the selected panel.

Managing nodes

To let users modify or delete nodes already present in the graph, each item in the Workspace has two buttons for deleting and editing the node, respectively.

Clicking the delete button (red background with a trash icon) and confirming in the browser alert deletes the selected node and any relationship connected to it.

Clicking the edit button (white background with a pencil-over-square icon) opens a modal dialog with the selected node's information, from which you can make changes and then save them.

Some items also have a third button. These items are stratigraphic units associated with semantic shapes. This button serves two purposes:

- Identify nodes that have an associated semantic shape
- On click, make the semantic shape visible if it is present in the scene.

Shelf panel

It is one of the key components for enriching the knowledge graph semantically, as it serves as the information source to associate with stratigraphic nodes. It contains 3D models, images, and documents that can be added to existing graphs in two ways: by dragging items into the Workspace panel or directly into the scene.

New content can be added to the Shelf via dedicated buttons in the sections for each resource type. You can also filter the contents here using a search bar.

Advanced search

Using the search bar, you can find the desired content within the active panel. If you start typing text, the system shows items whose names contain the entered text; if you enter a resource URL, the system will show, if present, the item whose URL matches what you entered. Since GLTF files come with an accompanying BIN file and a PNG texture, providing the GLTF file link is sufficient to find the item.

Adding items to the shelf

Using the buttons in the specific sub-panels, you can enrich the Shelf's contents. New items are created via a modal dialog where you can specify:

- Item name
- Item description
- Remote link or local file(s) for the resource

As noted, you can add 3D models, images, and documents.

For uploading 3D models, you can use GLTF, GLB, OBJ, B3M (Cesium tileset), and FBX files. If you want to upload a GLTF file:

- For local files, you must also upload the associated BIN file and the PNG texture
- For remote links, the associated BIN file and PNG texture must reside in the same repository

Supported image formats:

- JPG
- PNG
- TIFF

Supported document formats:

- PDF
- DOC
- TXT

Shelf → Workspace

One way to add content is to drag an item from the Shelf onto a stratigraphic node in the Workspace ([Figure 5.3](#)). Unlike dragging into the scene — which is limited to 3D models — this method can be used with any type of content available on the Shelf.

After dragging, a panel opens, allowing you to choose the type of node to associate with the resource. Once you make this choice, the system automatically computes the sequence of nodes and relationships needed to link the two elements and opens a modal window ([Figure 5.4](#)) to finalize creation of the path.



Figure 5.3: Example of dragging from Shelf to Workspace

Shelf → Scena

This mode is reserved exclusively for 3D models and involves dragging the item from the Shelf onto any point of a model in the scene.

Once the drag operation is completed, two buttons become active in the scene controls panel: one lets you remove the object from the scene, and the other lets you actually add it to the graph. Adding it to the graph uses the same modal window described in the previous mode ([Figure 5.4](#)).

The 3D model placed in the scene can be adjusted in position, rotation, and scale using a gizmo, which is activated by clicking the object and selecting the desired mode (Figure 5.5). If multiple objects are present in the scene, the gizmo is associated with the currently selected one, making it the only target of the changes.

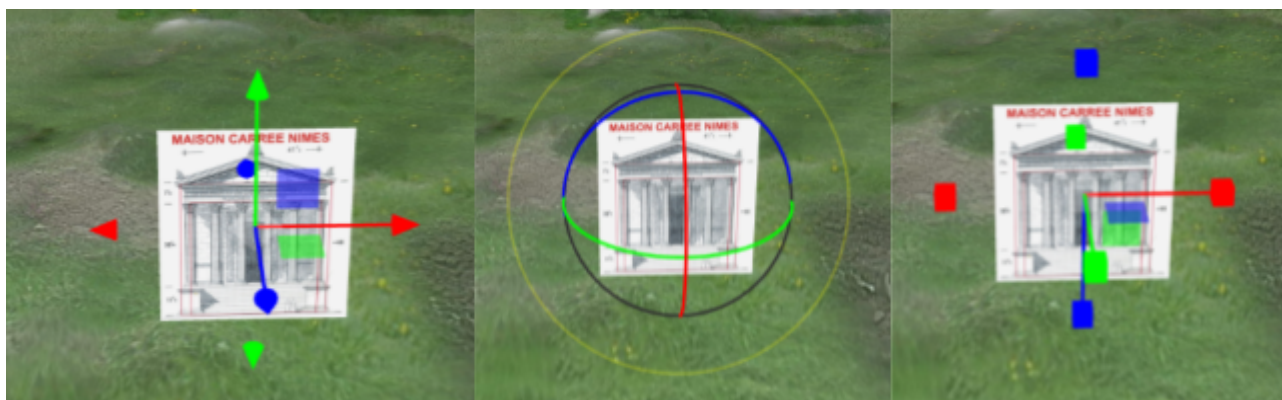


Figure 5.5: Gizmo modes

Automatic path creation

For the semantic enrichment performed using Shelf items, a specific procedure is used. After dragging into the Workspace or clicking the *Add to graph* button (found in the scene controls), a modal appears that lets you choose what type of element should be associated with the Shelf link. The options are:

- Representation Model;
- Document;
- Special Find.

A special case arises when adding images or documents, since they can be inserted into a graph only by linking them to a Document node; for this reason, in these two cases, only the *Document* option is available. For 3D models, instead, you can choose any of the three types:

- *Representation Model*, if you want to add it to a stratigraphic unit in general
- *Document*, if you want to avoid binding the model to epochs
- *Special Find*, if you want to link it to a special find.

After making this choice, the system uses an algorithm to automatically compute the shortest path that will connect the Shelf item and the stratigraphic node in the graph.

At the end of this procedure, the modal shown in [Figure 5.4](#) appears. There will be as many steps as the number of nodes involved in the path; for steps associated with known elements, the fields are auto-filled.

To guide the user through the steps, there is a sidebar structured like a graph, where each circle represents a step to complete. To provide feedback on completion status, each circle can appear in three colors:

- Gray, if no node has been selected and no information has been entered in the forms
- Orange, if the information has been entered only partially
- Green, if the information provided is sufficient.

✕



↓



↓



Seleziona nodo Link

Seleziona un elemento...

Nome *

Temple Column

Descrizione *

Doric column with base and capital

Tipo *

3d_model

File Locali *

Scegli file Nessun file selezionato

File Remoto *

https://dataspace.ispc.cnr.it/models/temple_column.glb

Next

Figure 5.4: Example of a modal for creating an automatic link

Tool panel

Within this panel, several buttons are available for creating nodes and relationships. Each button opens a dedicated panel where you can enter all the information needed to define the element you want to create.

Creating semantic shape

This panel also includes two buttons that enable the semantic mask drawing mode. To start, click the **Draw a semantic shape** button. From that point, you can place points to enclose the semantic shape; you can create both 2D shapes and polygons. When you're satisfied with the points you've placed, click **Finalize path**, which opens a modal where you can enter:

- A description to assign to the shape

- The stratigraphic node to link it to.

After clicking **Done**, the system automatically creates the node associated with the mask and the relationship linking it to the stratigraphic unit.

Scene Controls panel

In addition to the brightness control and the list for selecting which relationships to draw — already available in the viewer — a set of buttons is provided to:

- Save the graph state
- Manage the objects added from the Shelf to the scene
- Export the graph and the desired semantic shapes

Saving the graph

This action is essential to make your changes permanent. When an operation is performed in the editor, it is not written directly to the file on the server. This lets you work on the graph without worrying about damaging the structure as it existed before the current session. To apply the changes to the graph as well, click the *Save changes* button. After using the button, the scene reloads, and you can verify that the changes have been permanently applied. If you do not save, the next time the page is reloaded, the graph reverts to the state resulting from the last saved modification.

Managing models from the Shelf

Once 3D models have been added to the scene, you can use two buttons in the panel to:

- Remove the model from the scene
- Add it to the graph's nodes

When a model is added to the scene, it is not added as a node in the graph. This way, you can reposition it in the scene and then decide whether to remove it using **Remove from scene** or add it to the graph as a node by clicking **Add to graph**.

Exporting the graph

By clicking **Export scene**, a JSON file describing the scene composition is downloaded. The exported graph reflects the scene as it currently exists, including changes that have not yet been committed on the server. The file is saved to the folder set as the default in your browser settings.

Exporting Semantic Shapes

Another feature lets you download the semantic masks present in the visible scene. This allows you to export masks created roughly in the Heriverse editor, import them into a modeling program for refinement, and then recreate the scene with the improved model. To export the masks, first click **Export Semantic Shapes**. A modal then appears with a selectable list; each item corresponds to a semantic mask present in the scene. After

selecting the items of interest, click **Export selected** to start downloading a compressed archive containing the corresponding model files.

Collaborative mode

Heriverse allows you to collaborate with other users within the same scene, in both viewing and editing modes.

During a collaborative session:

- Each user is represented by an avatar visible in the scene
- A text chat is available and can be exported at any time
- Voice communication can be enabled between participants
- Screen sharing is available: the shared content is projected onto a virtual plane visible in the scene. This plane can be moved, rotated, and resized as needed.

This feature is useful for co-design activities, group reviews, or guided presentations in immersive environments.

Starting a collaborative session

An authenticated user can start a collaborative session with others by clicking the “Start new session” button at the bottom-left of the screen. After starting the session, the chat box opens; you can copy the link using the dedicated button and invite other users simply by sending them the link. Users who receive the link can paste it into their browser to join the collaborative session.

Session chat

Users in the same session can send chat messages via the chat panel at the bottom-left of the screen. In addition to user messages, the chat box also displays system messages (for example: “User Mario Rossi has joined the session”). System messages are centered; messages received by the user are left-aligned; messages sent by the user are right-aligned. For messages sent by users, the sender’s initials (first and last name) appear next to each message.

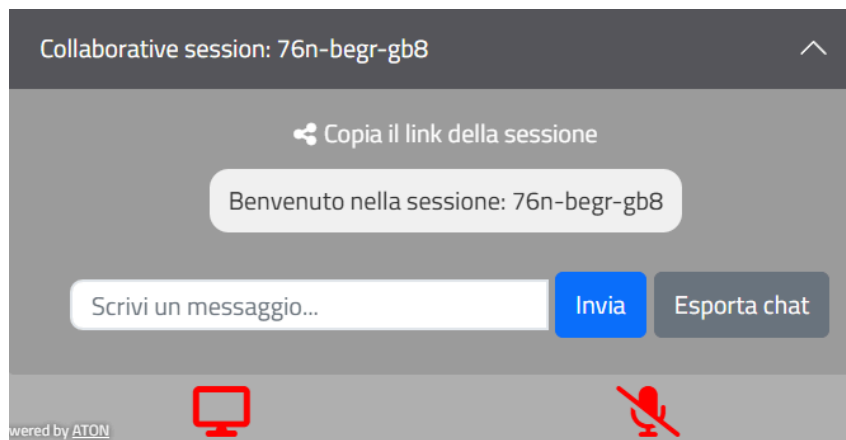


Figure 6: Shared chat and controls for screen sharing and voice chat

Session chat export

Users can keep track of what's written during a session by downloading the chat at any time. By clicking the **Export chat** button, a text file containing the entire chat log is downloaded. Each line in the file includes:

- The sender's initial in square brackets
- The actual message.

System messages are prefixed with [System].

Audio sharing

Users can start a voice chat session by clicking the microphone icon in the chat panel. Audio is spatial within the scene, so the closer users are, the louder it sounds; if avatars are far apart, the audio will be heard at a distance. Users can end the voice chat by clicking the microphone icon again.

Screen sharing

After starting — or joining — a collaborative session, users can share their screen with others. To start sharing, click the camera icon in the chat panel and choose what to share (screen, tab, application). All users will see the shared screen within the 3D scene. The user who initiated sharing can move, rotate, or scale the view using the available gizmos by clicking the screen panel in the 3D scene. Other participants will see the screen move or resize in real time.

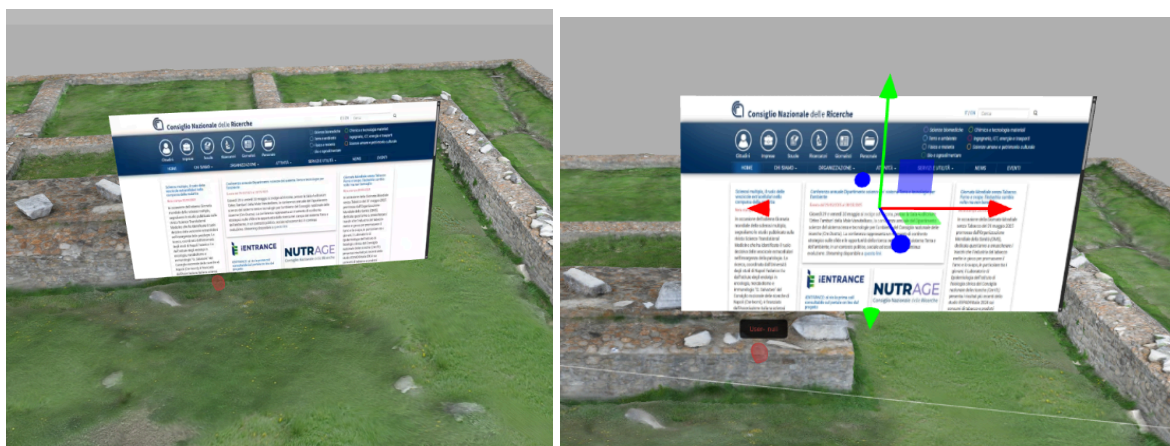


Figure 6.1: Example of Screen sharing